Assignment 1: EC5644

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1 Question 1

The first question is premised upon the below Gaussian distribution, which identifies a four-dimensional vector space with the related means and covariances.

$$\mathbf{m}_0 = \begin{bmatrix} -1/2 \\ -1/2 \\ -1/2 \\ -1/2 \end{bmatrix} \quad \mathbf{C}_0 = \frac{1}{4}\begin{bmatrix} 2 & -0.5 & 0.3 & 0 \\ -0.5 & 1 & -0.5 & 0 \\ 0.3 & -0.5 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} \quad \mathbf{m}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \mathbf{C}_1 = \begin{bmatrix} 1 & 0.3 & -0.2 & 0 \\ 0.3 & 2 & 0.3 & 0 \\ -0.2 & 0.3 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

1.1 ERM Classification Rule

$$\begin{aligned} D_{\text{ERM}}(x) &= \arg\min\, R(D = d \mid x) \\ &= \arg\min\, (\lambda_{d0}\, P(L = 0 \mid x) + \lambda_{d1}\, P(L = 1 \mid x)) \end{aligned}$$

Decide $D(x) = 1$ when $Risk(D = 1 \mid x) < Risk(D = 0 \mid x)$:

$$\frac{P(L = 1 \mid x)}{P(L = 0 \mid x)} > \frac{\lambda_{10} - \lambda_{00}}{\lambda_{01} - \lambda_{11}}$$

Apply Bayes Rule:

$$\frac{P(x \mid L = 1)}{P(x \mid L = 0)} > \frac{(\lambda_{10} - \lambda_{00})\, P(L = 0)}{(\lambda_{01} - \lambda_{11})\, P(L = 1)}$$

Assume that making a wrong decision costs more than making a correct decision.
Given: $\lambda_{10} = 1,\ \lambda_{00} = 0,\ \lambda_{01} = 1,\ \lambda_{11} = 0$

$$\frac{P(x \mid L = 1)}{P(x \mid L = 0)} > \frac{(1)\, P(L = 0)}{(1)\, P(L = 1)}$$

$$\gamma = \frac{0.35}{0.65}$$
$$\gamma = .538$$

1.2 ERM Classification

Utilizing an ERM classifier, we can estimate the error a variety of thresholds and calculate the True Positive Rate and False Positive Rate for each γ, building and approximate ROC curve. To decide an optimal γ, this can be done empirically or theoretically with decision rule derived in section 1.1. Figure 1 depicts the associated ROC Curve, with labeled points of the associated theoretical minimum and the empirically derived minimum. Through this process, the **theoretically best γ of .538** produced an **error of 0.0267**. Empiricaly, a threshold of γ **= 0.62** was derived, yielding an **error of 0.0257**.

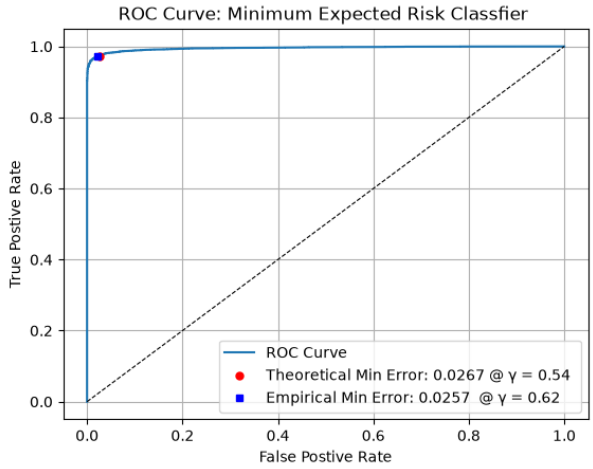


Figure 1: ERM Classifier ROC Curve and Minimized Error.

1.3 Naive Bayes Classification

Naive Bayes classification assumes features are independent given each class label, meaning the covariance matrices of each Gaussian distribution would be diagonal (equal to true variances for each non-zero value in the covariance matrix). This method achieved a higher total error due to the mismatch between the classification method and naive Bayes assumptions versus the reality of the true distribution. These plots are reported in Figure 2 as well as Figure 3, which depicts the ROC curves for both the Naive Bayes Classifier as well as the comparison between the Naive Bayes Classifier and the ERM Classifier. The weakness of Naive Bayes in this circumstance was the incomplete information it utilized in not assuming covariance between the features, leading to a lower best error, derived empirically of **0.0315 at the γ threshold of .77**.

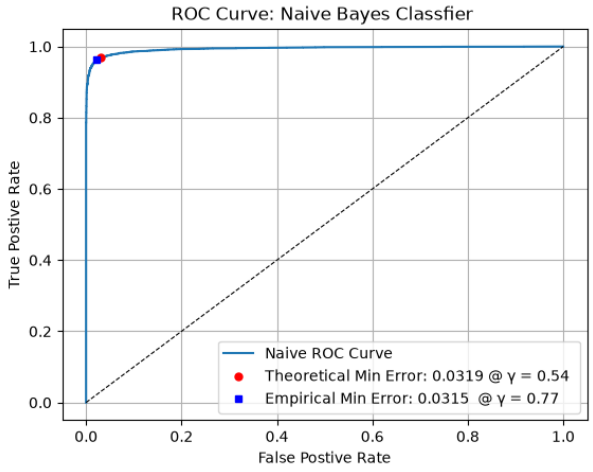


Figure 2: Naive Bayes classifier ROC curve and minimum-error operating points.

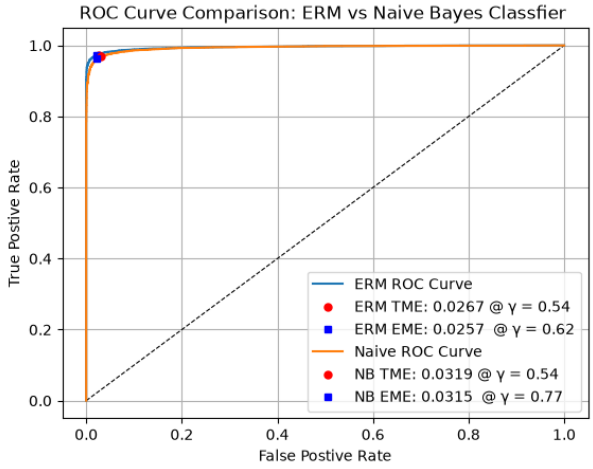


Figure 3: Comparison of the ERM and Naive Bayes classifiers.

1.4 Fisher LDA

Fisher Linear Discriminant Analysis performs a project of a given distribution onto a 1-dimensional line, creating a decision boundary for classification between two classes. This decision boundary is found through the derivation of the S_B, the **Between Class Scatter** and S_W, the **Within Class Scatter**. These

equations are as follows:

$$S_W = \hat{\Sigma}_0 + \hat{\Sigma}_1 \tag{1}$$

$$S_B = (\hat{\mu}_1 - \hat{\mu}_0)(\hat{\mu}_1 - \hat{\mu}_0)^T \tag{2}$$

$$S_B\, w_{LDA} = \lambda\, S_W\, w_{LDA} \tag{3}$$

These equations describe, as the names suggest, the separation between the class means and the spread within each class around its own mean, respectively. These two terms combine in Equation 3, which describes the vector w_{LDA}: the generalized eigenvector of S_B and S_W with the largest associated eigenvalue, λ. Using this eigenvector, each sample x_n is assigned a score equal to its projection onto w_{LDA}, $(w_{LDA})^T x_n$, which is compared against a threshold τ. Since τ has no closed-form optimal value, it is found by sweeping candidate thresholds from $-\infty$ to ∞ and selecting the one that minimizes the empirical probability of error. The resulting decision boundary, $(w_{LDA})^T x = \tau$, is a hyperplane. This sweep and optimal value, τ, is described in the below ROC graph, as well as a comparison to the prior classification methods of ERM and Naive Bayes Classification. Fisher LDA performed behind ERM and Naive Bayes, yielding an **error of 0.0489 at the τ threshold of .03**.

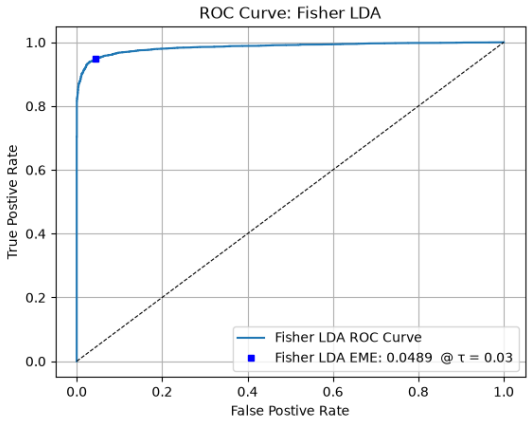


Figure 4: Fisher LDA classifier ROC curve and minimum-error operating points.

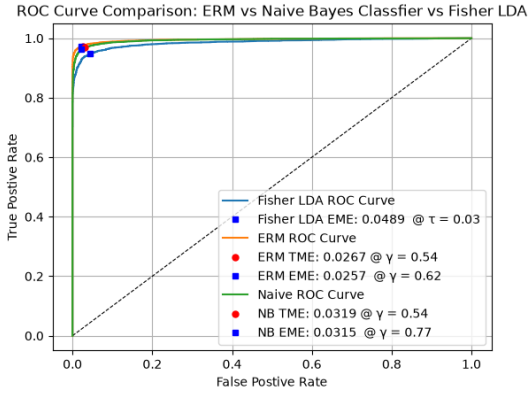


Figure 5: Comparison of the ERM, Naive Bayes, and Fisher LDA Classifiers.

1.5 Classifier Comparison

Though this exercise, we have derived three distinct classifiers, those being ERM Classification, Naive Bayes Classification, and Fisher Linear Discriminant analysis. As seen in both Figure 5 as well as in Table 1, the lowest error, and thus highest performance was seen in the ERM Classification, followed by the Naive Bayes, and the lowest with Fisher LDA. For the performance of each of these metrics, their relative performance follows intuitively. ERM's basic assumption is that the data exists in some Gaussian Distribution with some μ and Σ which can be leveraged to create a classification rule. This perfectly matches the data, making this an effective classification method.

For Naive Bayes, this classification method assumes a diagonal Covariance matrix, which was incompatible with the true distribution. However, the diagonalized Covariance matrix was a close enough match to the original distribution to allow for effective classification, but was still dominated by the ERM classification.

As for Fisher LDA, this methodology was the least effective of the chosen classification methods. The root of this issue stems from the optimal case for Linear Discriminant Analysis is when the Covariance Matrices of the given distributions are equal, or generally equal. When finding the sum of the variance of the given distributions, found through np.trace, $trace(C_0) = 1.51$ and $trace(C_1) = 7.00$, with the variance of $g(x|m_1, C_1)$ being 4.7 times greater than $g(x|m_0, C_0)$. This significantly contributes to the incompatibility of Fisher LDA for this classification problem.

Classifier	Threshold selection	Threshold	P(error)
ERM (true pdf)	Theoretical (TME)	$\gamma = 0.54$	0.0267
	Empirical (EME)	$\gamma = 0.62$	0.0257
Naive Bayes	Theoretical (TME)	$\gamma = 0.54$	0.0319
	Empirical (EME)	$\gamma = 0.77$	0.0315
Fisher LDA	Empirical (EME)	$\tau = 0.03$	0.0489

2 Question 2

2.1 Data Distribution

The specifications for the distributions in this problem were for each μ of the gaussian distributions to be equidistant with large enough covariance matrices as to allow for some overlap between class labels. With this guidance, the following gaussian distributions were created and sampled 10,000 times to create a list of the labeled data. The mean vectors are

$$\mathbf{m}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{m}_2 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}, \quad \mathbf{m}_3 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{m}_4 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix},$$

The covariance matrices are

$$\mathbf{C}_1 = \begin{bmatrix} 2 & -1.5 & 0.5 \\ -1.5 & 2 & 1 \\ 0.5 & 1 & 5 \end{bmatrix}, \quad \mathbf{C}_2 = \begin{bmatrix} 1 & -0.2 & -0.5 \\ -0.2 & 2 & 0.5 \\ -0.5 & 0.5 & 1.5 \end{bmatrix}, \quad \mathbf{C}_3 = \begin{bmatrix} 1 & 0.1 & 0.5 \\ 0.1 & 1.5 & 1 \\ 0.5 & 1 & 1.5 \end{bmatrix}, \quad \mathbf{C}_4 = \begin{bmatrix} 2 & 0.1 & 0.5 \\ 0.1 & 2 & 0.1 \\ 0.5 & 0.1 & 0.5 \end{bmatrix}.$$

These four gaussians are translated into three classes with associated priors, described by the following equations and visualized in Figure 6.

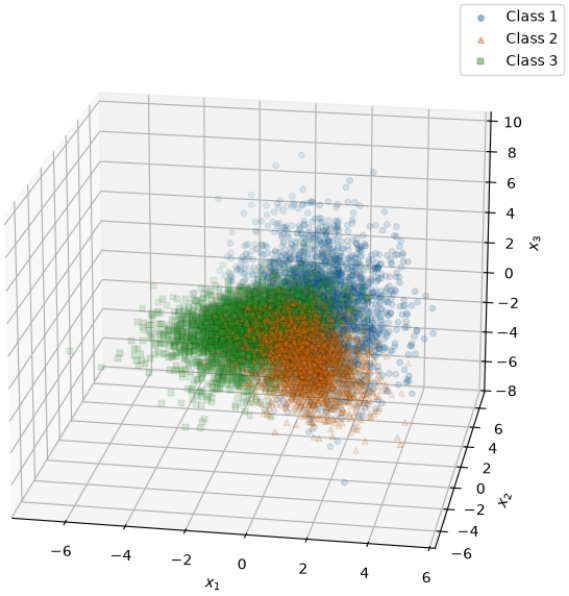


Figure 6: True Labels of Classes 1, 2, and 3.

$$p(\mathbf{x} \mid L = 1) = g(\mathbf{x} \mid \mathbf{m}_1, \mathbf{C}_1), P(L = 1) = 0.3$$
$$p(\mathbf{x} \mid L = 2) = g(\mathbf{x} \mid \mathbf{m}_2, \mathbf{C}_2), P(L = 2) = 0.3$$
$$p(\mathbf{x} \mid L = 3) = \tfrac{1}{2}\, g(\mathbf{x} \mid \mathbf{m}_3, \mathbf{C}_3) + \tfrac{1}{2}\, g(\mathbf{x} \mid \mathbf{m}_4, \mathbf{C}_4), P(L = 3) = 0.4$$

2.2 Classification

Since the true class-conditional pdfs and class priors are known, no parameters are estimated in this part: the classifier uses the exact means, covariance matrices, mixture weights, and priors from Section 2.1. To minimize the probability of error, the 0-1 loss matrix is used, which assigns zero cost to correct decisions and unit cost to every incorrect decision:

$$\Lambda = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}. \tag{4}$$

The ERM rule selects the decision with the smallest conditional risk, $R(D = i \mid \mathbf{x}) = \sum_j \lambda_{ij} P(L = j \mid \mathbf{x})$. Under 0-1 loss, $R(D = i \mid \mathbf{x}) = 1 - P(L = i \mid \mathbf{x})$, so minimizing the risk is equivalent to choosing the class with the largest posterior. Applying Bayes' rule and dropping $p(\mathbf{x})$, which is common to all classes, gives the minimum-probability-of-error (MAP) decision rule:

$$D(\mathbf{x}) = \arg \max_{k \in \{1,2,3\}} \; p(\mathbf{x} \mid L = k)\, P(L = k) \tag{5}$$

This rule was applied to all 10,000 samples, producing an error of 0.2129. This classification is described below in Figure 7.

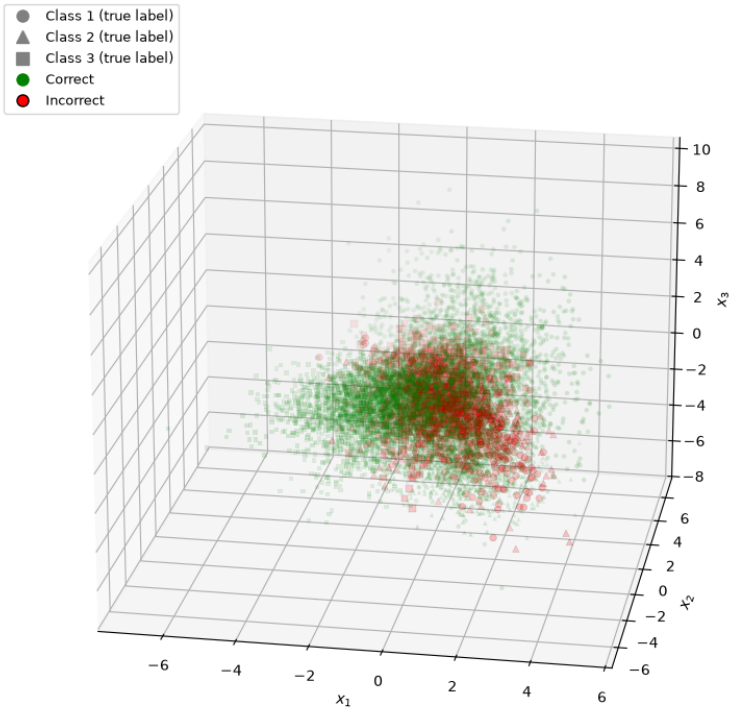


Figure 7: Classification Errors. P(Error) = .2129 | 2129 total errors

2.3 Classification With Exaggerated Loss Matrices

With a loss matrix, it can be augmented that punish mistakes from a particular class. With the below loss matrices, they increase the error of misclassifications of class three by a factor 10 and 100, described below in Equation 7.

$$\Lambda_{10} = \begin{bmatrix} 0 & 1 & 10 \\ 1 & 0 & 10 \\ 1 & 1 & 0 \end{bmatrix} \qquad \text{and} \qquad \Lambda_{100} = \begin{bmatrix} 0 & 1 & 100 \\ 1 & 0 & 100 \\ 1 & 1 & 0 \end{bmatrix} \tag{6}$$

As can be expected, the classifier begins to preference class label 3 at a higher preference. This results in a corresponding decrease in the general accuracy of the model and a corresponding decrease in the error of Class 3. These findings are highlighted below in Table 2 and the corresponding confusion matrices are shown in Figure 8.

Loss matrix	Class 1	Class 2	Class 3	Errors	P(error)
0-1 loss (MAP)	0.286	0.221	0.151	2129	0.213
Λ_{10}	0.595	0.392	0.015	3043	0.304
Λ_{100}	0.796	0.619	0.001	4289	0.429

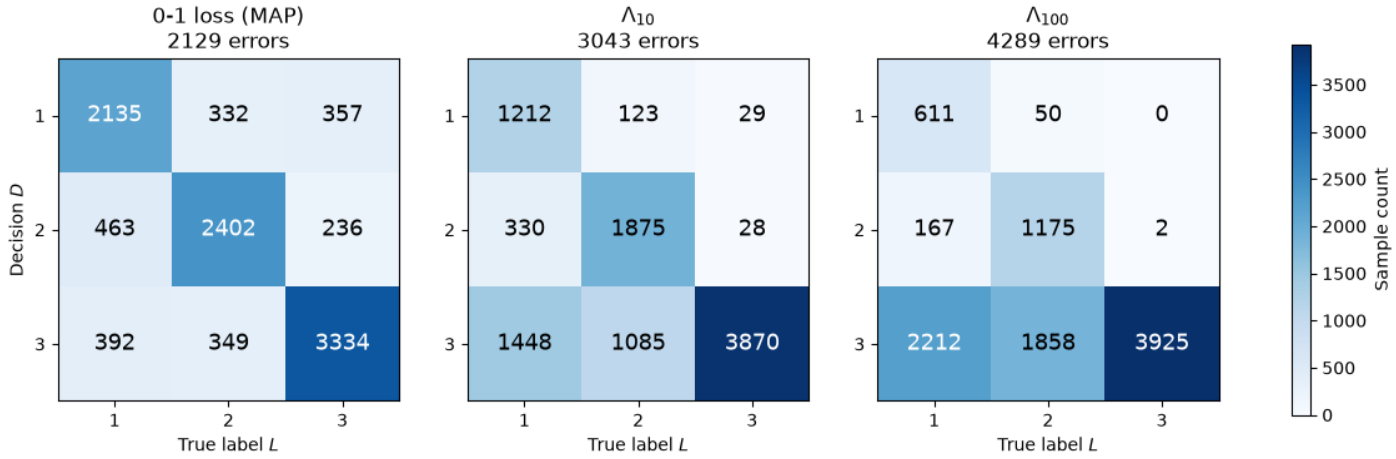


Figure 8: Confusion matrix comparison between loss functions.

2.4 Conclusion

This question effectively highlight the significance of loss functions and their corresponding effect on accuracy of individual class of distributions. Crucially, though these modifications of the loss matrices overall lowered the overall accuracy of the model, the risk of misclassifying $L = 3$ decreases significantly, which can be advantageous for applications such as healthcare where we would want to minimize the risk of certain patients having a "missed" diagnosis to receive additional testing, as the risk of missing a serious illness outweighs cost of a few more tests.

3 Question 3

3.1 Wine Quality Dataset

The white wine quality dataset consists of 4898 samples with 11 features and 7 distinct wine quality scores (3 through 9). Each class-conditional pdf was modeled as a multivariate Gaussian; this is a modeling assumption whose suitability is discussed in Section 3.3. Since the features have very different scales, each feature was first standardized to zero mean and unit variance. Class priors were estimated from the relative sample size of each class, and the means and covariance matrices of each class were estimated with sample averages (maximum likelihood estimation). Regularization was necessary because the rarest class, quality 9, contains only 5 samples in 11 dimensions, so its sample covariance has rank at most 4 and cannot be inverted. Each sample covariance was therefore regularized by adding $\lambda \mathbf{I}$, where λ equals α times the trace of the sample covariance divided by its rank, with $\alpha = 0.01$:

$$\hat{\mathbf{C}}_k = \hat{\mathbf{C}}_k^{\text{sample}} + \lambda_k \mathbf{I}, \qquad \lambda_k = \alpha \, \frac{\operatorname{tr}(\hat{\mathbf{C}}_k^{\text{sample}})}{\operatorname{rank}(\hat{\mathbf{C}}_k^{\text{sample}})}. \tag{7}$$

After regularization, all seven covariance matrices were full rank. Using these estimates, two classification rules were applied: Maximum A Posteriori (MAP) and Maximum Likelihood (ML) classification. MAP had an error rate of 0.505 and ML had an error rate of 0.623. The corresponding confusion matrices are shown in Figure 9.

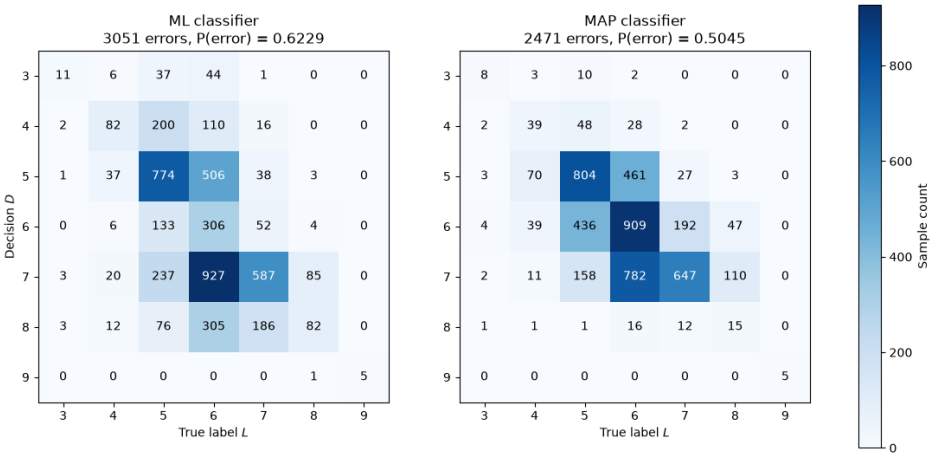


Figure 9: Wine Quality Classification confusion matrix comparison between ML and MAP.

3.2 Human Activity Recognition Dataset

UCI's Human Activity Recognition dataset contains 6 classes of different activities such as walking, sitting, standing, and laying described through accelerometer and gyroscope data. Classification of this dataset was done with the same methodology as described in the Wine Quality dataset in Section 3.1. Parameters were estimated through MLE and classified through MAP and ML. The results of these classifications are described in the confusion matrices in Figure 11. The classification techniques utilized proved particularly effective, with a ML error of 0.0367 and MAP error of .0369. The main source of confusion came from sitting examples being classified as standing, which is logical due to the stationary data looking very similar through the instruments utilized in the study.

3.3 Dataset Comparison

Comparing the two datasets, the UCI Human Activity Recognition Study and the Wine Quality Dataset, key differences can be characterized as class balance and data differentiability. Observing the class balance

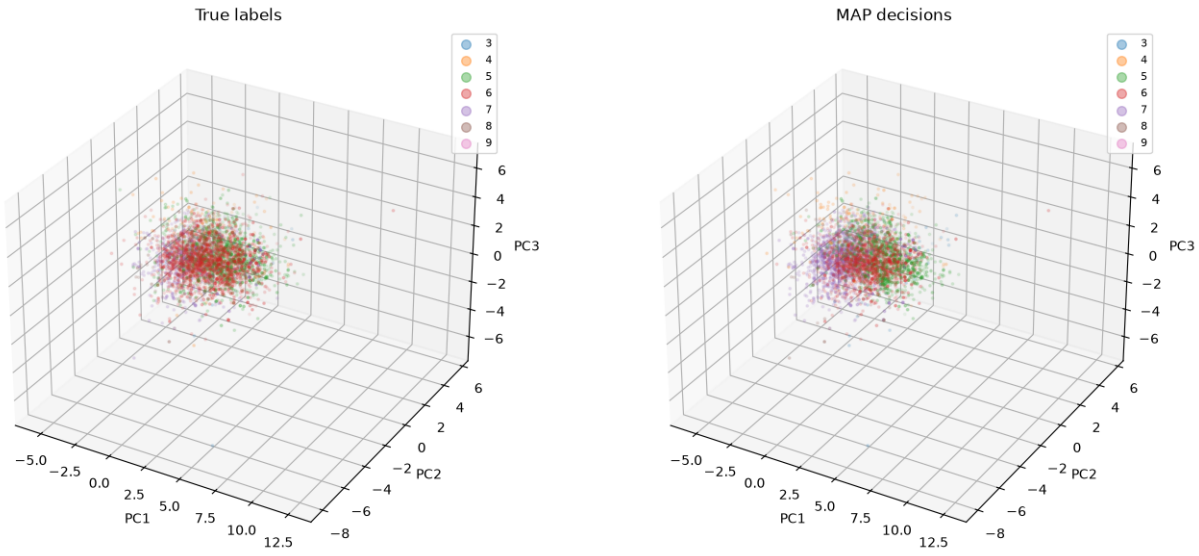


Figure 10: PCA and Classification of Wine Quality Dataset

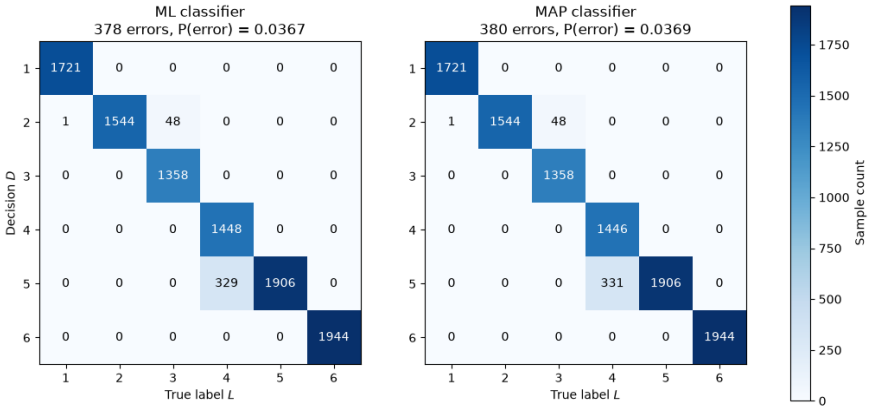


Figure 11: UCI HAR Classification confusion matrix comparison between ML and MAP.

of the HAR dataset compared to the wine quality dataset, it can be seen that HAR has approximately even class distributions. Furthermore, the Wine Quality dataset lacks a signifigant amount of samples per class, which has as little as 5 samples in class 9. The fundamental difference between the two rules is the log-prior term $\ln \hat{P}(L = k)$, which MAP includes and ML omits. This term depends on how unequal the class priors are, not on the number of samples: it does not vanish as N grows. When the classes are imbalanced, as in the wine dataset where qualities 5 and 6 make up roughly three-quarters of the samples, the prior term shifts decisions toward the common classes and MAP achieves a substantially lower error (0.505 vs. 0.623). When the classes are nearly balanced, as in HAR, the prior term is almost the same for every class, and MAP and ML make nearly identical decisions (0.0367 vs. 0.0369).

Based upon the performance of these models, it can be inferred that the use of gaussian distributions to estimate model parameters was more appropriate for the HAR dataset as opposed to the wine quality dataset. The Wine Quality dataset's highly skewed and, at times, incredibly sparse class labels made the dataset incompatible with the utilized gaussian models, explaining the poor performance in this case. On the other hand, HAR's differentiable and balanced design made the gaussian assumptions much more appropriate for this model.

Figures 10 and 12 show the first three principal components of each dataset. The HAR classes form

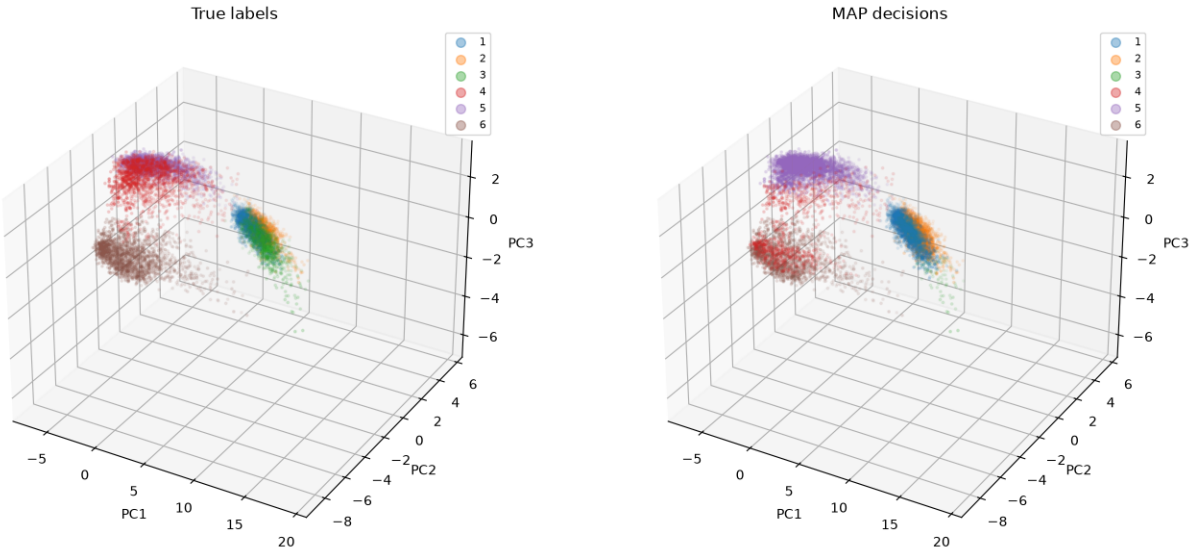


Figure 12: PCA and Classification of the UCI Human Activity Recognition Dataset

well-separated clusters, with moving activities (1–3) clearly apart from stationary ones (4–6), while the wine classes overlap almost completely. The first three components also capture more of the variance for HAR (.71) than for wine (.54). This separation comes from the features themselves: physical activities produce very different accelerometer and gyroscope signals, whereas wines of neighboring quality scores have similar chemical profiles. This overlap is the main reason the wine classifiers' error remains high regardless of the decision rule.

3.4 Conclusion

A key takeaway from this comparison is the idea of appropriate models for a given dataset. Even with regularization terms and tuning of parameters, the inherent non-gaussian nature of the Wine Quality dataset made it significantly under perform when compared to HAR. Such considerations are important when approaching classification problems and the optimization of the solutions.

Acknowledgment of AI Assistance

Claude (Anthropic), an AI assistant, was used throughout this assignment. Claude explained the underlying theory, including expected risk minimization and the likelihood-ratio test, the distinction between MAP and ML classification and maximum likelihood parameter estimation, Fisher LDA and the generalized eigenvalue problem, Naive Bayes, ROC curves, and covariance regularization. It pointed to relevant Python libraries and documentation and reviewed my code, identifying several bugs, including misaligned labels after sampling, sorted scores breaking the alignment with labels in the LDA classifier, swapped ROC axes, comparing log-likelihood ratios to γ instead of $\ln \gamma$, an incorrect loss matrix entry, and a missing scaling factor α in the HAR regularization. Claude also wrote portions of the code, including the loss-matrix comparison script, the confusion-matrix and PCA visualization functions, the class-balance script, and cleaned-up versions of the ROC and summary-statistics code; I integrated, ran, and adapted this code, and all reported numerical results were produced by running it on my generated samples and the downloaded datasets. In addition, Claude provided LaTeX formatting for equations, tables, and figures, corrected errors in my derivations, drafted or rewrote portions of the report text (including parts of Sections 1.4, 2.2, 3.1, 3.2, and 3.3), and reviewed drafts against the assignment requirements. These contributions significantly contributed to troubleshooting, understanding, and building a more comprehensive and higher quality assignment. Rewriting, coding, and evaluating this work was part of the key task I used Claude to accomplish. The transcript for the entire session can be found here: `https://claude.ai/share/51213d8b-5376-48f0-afff-951951cf182f`

All code used in this project, as well as figures, can be found at this Github link:
`https://github.com/RonanEngel/EC5564_Assignments`